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Concept Review

Passive Infrared Sensors

Why Do We Need Passive Infrared Sensors?

Ever entered a room and have lights turn on automatically? Or ever play a game that detects when you're stationary versus in motion? To accomplish these tasks electromechanical systems can make use of passive infrared sensors (PIR) to detect change in infrared emissions. PIR sensors fall within the family of motion sensors and can be used in a wide range of applications.

Passive Infrared Sensors

Passive infrared sensors, like the name suggests, utilize material properties of piezoelectric ceramic materials to detect changes in infrared emission. Typically, PIR sensors combine both a sensor plat and a lens to focus the infrared emissons. Figure 1 is an example of a possible sensor and lens combination for the Kemet PL- Q873.

a. Sensor front view

b. Sensor neutral lens

Figure 1: elecromechanical elements of the Kemet PL-Q873 PIR sensor

Another element to consider for PIR sensors is the field of view for which the lens can help focus infrared emissions.

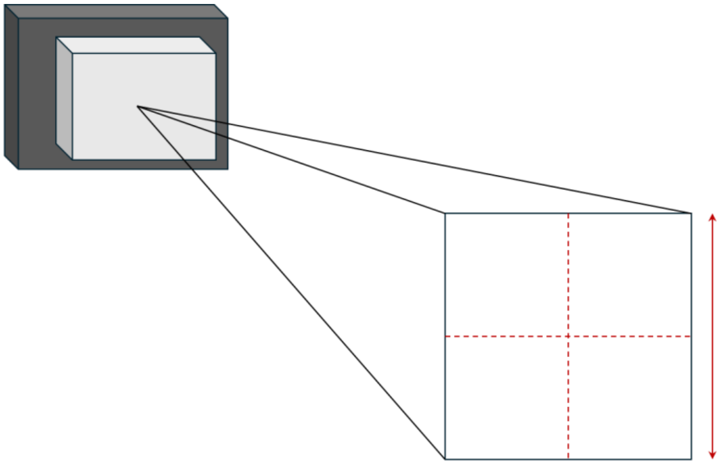
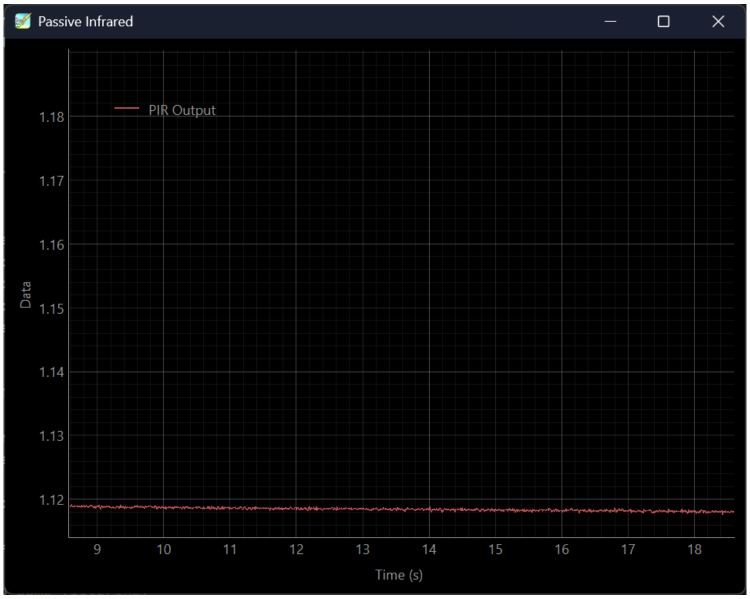
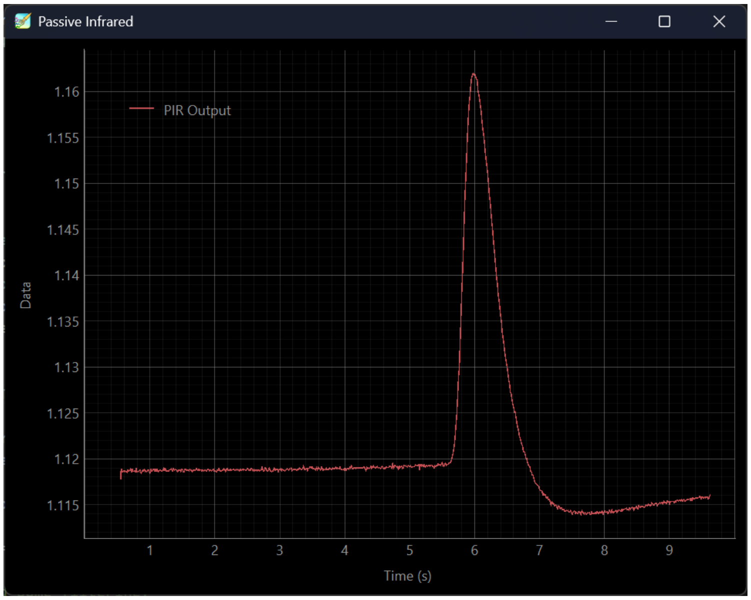


Figure 2: Representation of Passive Infrared sensor field of view

Sensor readings are measured in voltage and fluctuate depending if an object is moving withing the PIR sensor's field of view. Figure 3 is a representation of the PL-Q873 comparing a static environment versus motion of an object crossing the field of view of the sensor.



a. PIR measureent of static environment



b. PIR measurement of motion detected

Figure 3. Example voltage measurement of a PIR sensor.

The rate at which the PIR sensor voltage rises and falls is related to the sensitivity of the ceramic sensor. Likewise, since PIR sensors measure infrared emission, the change in temperature between the ambient environment and the object present has an impact on how consistently a voltage spike can be detected.